

INDIAN STANDARD SPECIFICATION: IS 1786 : 2008 (AMENDED)

HIGH STRENGTH DEFORMED STEEL BARS AND WIRES FOR CONCRETE REINFORCEMENT (TMT)

Standard Number: **IS 1786 : 2008** (Reaffirmed 2023). Mandatory under the Steel and Steel Products (Quality Control) Order. Covers Thermo-Mechanically Treated (TMT) bars for civil construction.

MECHANICAL PROPERTY REQUIREMENTS

Steel Strength Grade	Min 0.2% Proof Stress (MPa)	Min Tensile Strength (MPa)	Min Elongation (%)	Total Elongation at Max Force (TS/YS Ratio)
Fe 415	415.0	485.0 (or 1.10 x YS)	14.5%	Not specified
Fe 500	500.0	545.0 (or 1.08 x YS)	12.0%	Not specified
Fe 500D (Ductile)	500.0	565.0 (or 1.10 x YS)	16.0%	Min 5.0% (Earthquake Resistant)
Fe 550D	550.0	600.0 (or 1.08 x YS)	14.5%	Min 5.0%
Fe 600	600.0	660.0 (or 1.06 x YS)	10.0%	Not specified

MANDATORY MARKING ON BARS

Every metre of rebar must have rolled-on manufacturer brand name, grade (e.g., Fe 500D), diameter (e.g., 12mm), and the authentic BIS ISI Standard Mark.